

LEAD RESEARCH AGENT

Research, Architecture and Technical Design

AI Sales Co-Pilot — Agent 2 of 6

Version 1.0

Purpose

This document explains what the Lead Research Agent does, how it works, what data it needs, how it checks information, and how it passes a clean research result to the next sales agent.

1. What is a Lead Research Agent?

A Lead Research Agent is an AI system that researches a sales lead before outreach. It identifies the person and company, collects useful information, finds recent business signals, checks the lead against the Ideal Customer Profile (ICP), and returns a structured result.

The agent should not simply generate a paragraph about a lead. It should produce evidence-backed facts that another agent can safely use.

2. Problem

- Salespeople spend too much time collecting information manually.
- Lead records can be incomplete, duplicated or outdated.
- Person and company information is spread across many systems.
- Recent events such as funding, hiring and leadership changes are easy to miss.
- Wrong or unverified information can create bad personalization.
- Different sources can disagree, so the agent needs a way to handle conflicts.

3. Main Objectives

- Identify the correct person and company.
- Collect current person and company information.
- Verify important contact information.
- Find useful buying signals.
- Keep the source and confidence of important facts.
- Calculate an understandable ICP score.
- Apply suppression and safety rules.
- Return a clean Lead Research Package.

4. System Architecture

The agent follows one clear flow. Research steps are grouped together, then their results are checked before the final score and output are created.

Lead Research Agent — Architecture

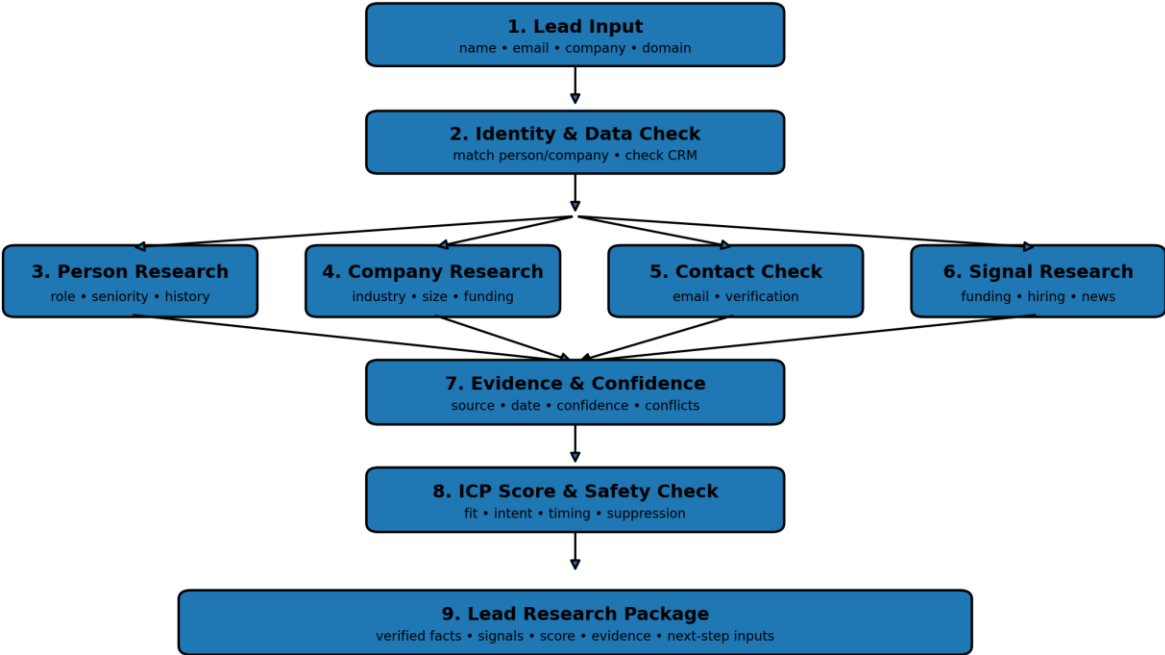


Figure 1. Lead Research Agent architecture.

5. How the Agent Works

- 1. Receive the lead.
- 2. Match the person and company.
- 3. Check existing CRM data.
- 4. Research person, company, contact and business signals.
- 5. Attach evidence and confidence to important facts.
- 6. Calculate ICP fit and lead priority.
- 7. Run the safety and suppression check.
- 8. Create the Lead Research Package.
- 9. Pass the package to the next sales agent.

6. Input Data

Input	Purpose
Lead ID	Find the lead in the internal system.
Name	Help identify the person.
Email	Identify or verify the professional contact.
Company	Find the correct organization.
Domain	Strong company identity signal.
Profile URL	Help distinguish people with similar names.
ICP / Campaign ID	Select the correct scoring rules.

7. Information the Agent Collects

Person information

- Name

- Job title and department
- Seniority
- Location
- Work history
- Professional contact details where permitted

Company information

- Company name and domain
- Industry
- Employee count
- Location
- Funding
- Leadership
- Hiring activity
- Technology
- Recent company activity

8. Buying Signals

Signal	Example	Why it matters
Funding	New funding round	May indicate growth and budget.
Hiring	Hiring for a target team	May indicate an active business need.
Leadership	New executive	Can change priorities.
Expansion	New market or office	Can create new requirements.
Product	New product launch	Can create new operational needs.
Technology	New relevant technology	May show a technical priority.
Company news	Major partnership	Provides current sales context.

9. Data Sources and Tools

The agent should access external systems through separate tool adapters. This keeps the main agent independent from any single provider.

Tool type	Use
CRM / database	Existing lead data, history and suppression status.
B2B enrichment	Person and company information.
Email verification	Find or verify professional email addresses.
Business data	Funding, hiring, leadership and company signals.
Approved web/news search	Recent public company information.
Internal knowledge	ICP, product and sales rules.

10. Waterfall Enrichment

Use the cheapest and most trusted available information first. Do not call every service when the needed field is already fresh and verified.

Lead → CRM → Primary enrichment → Fallback enrichment → Verification → Signal research → Final package

- Check internal data first.
- Use the primary provider for missing or stale fields.
- Use a fallback provider only when needed.

- Verify important contact details.
- Record which tool supplied each important value.

11. Evidence and Confidence

Important facts should be traceable. The agent must separate a fact from an AI-generated interpretation.

Field	Meaning
Claim	The information being reported.
Source	Where the information came from.
Date	When it was observed or retrieved.
Confidence	How reliable the information is.
Status	Verified, conflicting or unverified.

Example: “The company raised funding” is a fact and should have evidence. “The company may be expanding quickly” is an inference and should be labelled as such.

12. ICP and Lead Priority Score

Use fixed rules for the number. The AI can explain the result, but it should not freely invent the score.

Area	Example weight	Checks
ICP Fit	40%	Industry, size, location, role and technology.
Intent	30%	Funding, hiring, leadership and other signals.
Timing	20%	How recent and useful the signals are.
Confidence	10%	Identity and evidence quality.

Example priority levels: 80–100 High, 60–79 Medium, 40–59 Low, below 40 Do Not Prioritize. These values should be tuned using real sales results.

13. Lead Research Package

Section	Contains
Lead	Name, role, company and domain.
Company	Industry, size, location, funding and other verified data.
Signals	Important recent events.
Evidence	Source, date and confidence.
ICP	Fit, intent, timing and total score.
Compliance	Eligible, review required or blocked.
Personalization inputs	Approved facts for the content agent.

14. Agent State

State	Purpose
lead_input	Original lead request.
identity	Matched person and company.
person_profile	Person information.
company_profile	Company information.

contact_data	Email and verification status.
signals	Business signals.
evidence	Sources and confidence.
icp_score	Score and reasons.
compliance_status	Allowed, review or blocked.
research_summary	Final result.
errors	Problems during the run.

15. Recommended Technical Stack

Area	Technology
Backend	Python + FastAPI
Agent workflow	LangGraph
Validation	Pydantic
Database	PostgreSQL
Cache / queue	Redis when needed
Secrets	Environment variables or secret manager
Dashboard	Existing sales frontend

16. Security and Privacy

- Keep API keys and secrets on the backend.
- Use only data needed for the sales purpose.
- Restrict access to lead information.
- Keep an audit record of important research actions.
- Do not give the research agent permission to send emails.
- Do not allow unrestricted external requests.
- Treat external web content as untrusted input.
- Check suppression rules before outreach eligibility.
- Define retention and deletion rules for stored lead data.

17. Failure Handling

Problem	Action
Person cannot be matched	Ask for more information or send for review.
Provider unavailable	Retry within limits or use a fallback.
Conflicting data	Keep the conflict and lower confidence.
Missing data	Return unknown; never guess.
Low confidence	Send for human review when important.
Blocked lead	Do not pass the lead to outreach.
Invalid AI output	Validate and retry using the required schema.

18. Connection to Other Sales Agents

Agent	How it uses Lead Research
Context & Knowledge	Provides ICP, product information and rules.

Campaign Content	Uses verified research for personalization.
Outreach	Uses verified contact and eligibility information.
Inbox & Reply	Can provide new information for future research.
Lead Lifecycle	Uses research and score for lead progression.

19. MVP Scope

- Accept a person, company or domain.
- Resolve identity.
- Use one main enrichment service and one fallback.
- Verify a professional email when required.
- Find key business signals.
- Store evidence for important claims.
- Calculate an ICP score.
- Run the suppression check.
- Return the Lead Research Package.
- Log the research run and errors.

20. Implementation Order

10. Create Pydantic models.
11. Create the PostgreSQL tables.
12. Build identity resolution.
13. Build the first enrichment adapter.
14. Add email verification.
15. Add fallback enrichment.
16. Add business signal research.
17. Add evidence and conflict handling.
18. Add deterministic ICP scoring.
19. Add the safety gate.
20. Connect the workflow with LangGraph.
21. Connect the output to the Campaign Content Agent.
22. Test missing, conflicting and incorrect data.

21. Final Recommendation

The Lead Research Agent should be a controlled research workflow, not a free-running chatbot. Its main job is to collect useful information, verify it, find current business signals, score the lead and return a structured result.

The most important design rule is: the AI may summarize and explain information, but the system should control identity matching, validation, scoring, permissions, suppression and final state changes.